

The Generation-3 Spheroid-Guidance Model

Theory, Equations, and What Changed From Generation 2

Motor-clutch collagen-remodelling project / generations/g3_spheroid_guidance

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Abstract

中文摘要。 Generation 3 (**G3**, generations/g3_spheroid_guidance) 是一個二維機制示範模型，用來說明一顆癌細胞 **spheroid** (球體) 如何在沒有任何預先指定方向的情況下，自發打破對稱、伸出 protrusion (偽足) 跨越一圈「無纖維間隙」去抓住 collagen (膠原) 纖維、用 motor-clutch (馬達-離合器) 張力把周圍纖維網路往內拉並重排，最後往它真正抓得住的那一側遷移。**G3** 原封不動地重用了 Gloria 已驗證的 Generation 2 (**G2**) 膠原力學引擎 (generations/g2_corrected/common/model.py)：珠-彈簧網路、過阻尼動力學、軸向彈簧、離散彎曲、可自由轉動的永久 crosslink (交聯)、邊界錨定與細胞表面軟排斥全部不變。**G3** 只加了四樣東西：(1) 帶「無纖維間隙」的 spheroid 幾何；(2) 顯式 protrusion 與 tip-first (尖端先接觸) 綁定；(3) **emergent polarity** (湧現極性) ——一個質量守恆的 wave-pinning (波釘) 活性場加上 FAK/Rac1 式的黏著回饋，取代了 **G2** V3 寫死的 `polarity_probability = 0.65`；(4) 由反作用力驅動的二維遷移。本文用中英對照逐一寫出所有方程式、說明其運作，並明確標出哪些與 **G2** 相同、哪些被改動。

English. **G3** is a two-dimensional *mechanism demonstration*: it asks whether a cancer-cell *spheroid* can, with no prescribed direction, break symmetry, extend protrusions across an initially fibre-free gap to grip collagen, remodel the surrounding network by motor-clutch traction, and migrate toward the side where it actually grips. **G3** *reuses the frozen, validated G2 collagen engine unchanged* and adds only four ingredients: a spheroid with a fibre-free gap, explicit tip-first protrusions, an *emergent* wave-pinning polarity field with FAK/Rac1 adhesion feedback (which *replaces* the hardcoded `polarity_probability = 0.65` of **G2** V3), and reaction-driven 2-D migration. This report states every equation, explains how it works, and marks precisely what is shared with **G2** and what changed. *This is not a calibrated 3-D tumour-migration prediction.*

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1 Positioning and lineage / 定位與傳承

The project is organised as a sequence of “generations”. Each generation freezes the previous one and isolates a single new mechanism.

- **G2** (*corrected*) — a crosslinked, boundary-anchored bead–spring collagen network with overdamped dynamics, validated at $180\mu\text{m}$ / 99 fibres / 5 nN. Its sub-versions are **V2** (elastic crosslink transmission, fixed cell), **V3** (two-sided motor clutches + rigid-cell translation, $\sim 0.28\mu\text{m}/\text{min}$ migration), and **V4** (contact plasticity, a negative baseline).
- **G3** (*spheroid guidance*) — the subject of this report. It reuses the **G2** *engine* verbatim and adds a spheroid, protrusions, emergent polarity, and 2-D reaction-driven motion.

Why a rebuild. An earlier **G3** (`src/g3/`, now archived under `legacy/g3_v1_superseded/`) built a brand-new spatial-clutch engine from scratch. It was mis-scaled, ran for 15–60 s of simulated time — far too short for the tens-of-minutes-to-hours over which fibre realignment and migration actually develop — and produced chaotic animations that did not match **G2**. Its guidance test even *failed* (aligned nematic order $0.0142 < \text{isotropic } 0.0160$). The current **G3** discards that engine and instead stands on the already-validated **G2** mechanics.

Units. Throughout, lengths are micrometres (μm), forces are nanonewtons (nN), and time is seconds (s) — identical to **G2**, so the two models share one coordinate and force language.

2 The shared Generation-2 engine (unchanged) / 共用的 G2 引擎 (不變)

Everything in this section is *inherited from G2 without modification*. **G3**’s `SpheroidConfig` subclasses **G2**’s `CollagenConfig`, and **G3**’s integrator `fast_advance` reproduces **G2**’s `Network.advance` bit-for-bit (it only swaps `np.add.at` for `np.bincount` scatter-adds, $\sim 2.4\times$ faster). The collagen physics below is therefore common ground.

2.1 Overdamped bead force balance / 過阻尼力平衡

Collagen is a set of bead chains. Every mobile bead i obeys the professor’s overdamped (inertia-free) force balance

$$\zeta \dot{\mathbf{r}}_i = \mathbf{F}_i^{\text{stretch}} + \mathbf{F}_i^{\text{bend}} + \mathbf{F}_i^{\text{crosslink}} + \mathbf{F}_i^{\text{repulsion}} + \mathbf{F}_i^{\text{active}}, \quad (1)$$

where ζ is the bead drag (`bead_drag`). The viscous term $\zeta \dot{\mathbf{r}}_i$ dissipates motion into the surrounding fluid/porous matrix; the elastic bonds store mechanical energy. There is deliberately

no standard-linear-solid (SLS) element and **no** plasticity. The equation is integrated with an explicit forward-Euler step,

$$\mathbf{r}_i(t + \Delta t) = \mathbf{r}_i(t) + \Delta t \frac{\mathbf{F}_i^{\text{total}}}{\zeta} \quad (i \notin \text{fixed}), \quad \mathbf{r}_i \equiv \mathbf{r}_i^0 \quad (i \in \text{fixed}). \quad (2)$$

Boundary beads (those with $\max(|x_i|, |y_i|) \geq \frac{L}{2} - w_b$, where L is the domain size and w_b the boundary band width) are *fixed*: their positions are pinned to $\mathbf{r}_i^0$ so the network is anchored to a rigid outer frame. Stability requires $\Delta t < 2\zeta/k_t$; **G3** uses $\Delta t = 0.05$ s against $2\zeta/k_t \approx 0.32$ s.

2.2 Axial stretch / 軸向拉伸

Each bond (edge) joins consecutive beads i, j with initial rest length $\ell_0 = \|\mathbf{r}_j^0 - \mathbf{r}_i^0\|$. Writing $\mathbf{d} = \mathbf{r}_j - \mathbf{r}_i$, $\ell = \|\mathbf{d}\|$, extension $e = \ell - \ell_0$, the bond is a Hookean spring whose stiffness switches between tension and compression:

$$k = \begin{cases} k_t = EA/\ell_0, & e \geq 0 \quad (\text{tension}), \\ k_c = \rho k_t, & e < 0 \quad (\text{compression / microbuckling}), \end{cases} \quad \mathbf{F}_i^{\text{stretch}} = +k e \hat{\mathbf{d}}, \quad \mathbf{F}_j^{\text{stretch}} = -k e \hat{\mathbf{d}}. \quad (3)$$

Here EA is the axial rigidity (`axial_rigidity`), $EA = E A$ with fibre cross-section $A = \pi(D/2)^2$, diameter D , and modulus E . The compression ratio $\rho = 0.10$ (`compression_ratio`) softens compressed bonds, representing fibre microbuckling (Notbohm 2015). The stored energy is $U_{\text{stretch}} = \frac{1}{2} \sum k e^2$.

2.3 Discrete bending / 離散彎曲

For every interior triplet (a, b, c) of consecutive beads on a fibre, the discrete curvature is $\boldsymbol{\kappa} = \mathbf{r}_a - 2\mathbf{r}_b + \mathbf{r}_c$ with a stress-free reference $\boldsymbol{\kappa}_0$ taken from the initial geometry. Bending penalises the deviation $\Delta\boldsymbol{\kappa} = \boldsymbol{\kappa} - \boldsymbol{\kappa}_0$:

$$\mathbf{F}_a^{\text{bend}} = -k_b \Delta\boldsymbol{\kappa}, \quad \mathbf{F}_b^{\text{bend}} = +2k_b \Delta\boldsymbol{\kappa}, \quad \mathbf{F}_c^{\text{bend}} = -k_b \Delta\boldsymbol{\kappa}, \quad k_b = \frac{EI}{\bar{\ell}^3}, \quad (4)$$

where EI is the bending rigidity (`bending_rigidity`), $EI = E I$ with second moment $I = \pi D^4/64$, and $\bar{\ell}$ is the mean bond length. These forces are the negative gradient of $U_{\text{bend}} = \frac{1}{2} k_b \sum \|\Delta\boldsymbol{\kappa}\|^2$.

2.4 Freely-hinged permanent crosslinks / 可自由轉動的永久交聯

Where two *different* fibres cross, a permanent hinged link joins the two coincident material points. A material point on edge (a_1, a_2) at parameter α_a is $\mathbf{p}_a = (1 - \alpha_a)\mathbf{r}_{a_1} + \alpha_a\mathbf{r}_{a_2}$ (similarly $\mathbf{p}_b$). The link stores

$$U_x = \frac{1}{2} k_x \|(\mathbf{p}_b - \mathbf{p}_a) - \mathbf{d}_{x,0}\|^2, \quad \mathbf{f} = k_x [(\mathbf{p}_b - \mathbf{p}_a) - \mathbf{d}_{x,0}], \quad (5)$$

with $\mathbf{d}_{x,0} = \mathbf{0}$ for a permanent link (the two points are initially coincident) and stiffness k_x (`crosslink_stiffness`). The force $\mathbf{f}$ is distributed back onto the four host beads by the interpolation weights: $+(1 - \alpha_a)\mathbf{f}$ on a_1 , $+\alpha_a\mathbf{f}$ on a_2 , $-(1 - \alpha_b)\mathbf{f}$ on b_1 , $-\alpha_b\mathbf{f}$ on b_2 . The link is *freely hinged* (it resists separation of the two points, not the angle between fibres) and *never breaks* in **G2/G3** — there is no crosslink rupture manufacturing motion.

2.5 Soft cell-surface repulsion / 細胞表面軟排斥

The cell is a rigid circle of radius R centred at $\mathbf{r}_c$. Beads that penetrate its surface (plus a small clearance c) feel a one-sided linear penalty that pushes them radially out:

$$\delta_i = \max(0, R + c - \|\mathbf{r}_i - \mathbf{r}_c\|), \quad \mathbf{F}_i^{\text{repulsion}} = k_{\text{rep}} \delta_i \frac{\mathbf{r}_i - \mathbf{r}_c}{\|\mathbf{r}_i - \mathbf{r}_c\|}, \quad (6)$$

with $k_{\text{rep}} = \text{repulsion_stiffness}$ and $c = \text{cell_clearance}$. This is the repulsive branch of a linear-elastic contact; it adds no adhesion and no friction.

2.6 Network generation and the percolation gate / 網路生成與滲流閘

Finite curved fibres of length 20–110 μm are laid down (a sine-perturbed segment, function `_curved_segment`, reused from [G2](#)). Interior fibres are seeded at random angles; a fraction are “boundary-seeded” so they reach the anchored frame. Permanent crosslinks are then placed at all pairwise fibre intersections (`build_crosslinks`). A generated network is **accepted only if** it percolates: at least a required fraction of fibres are connected through the crosslink graph to the fixed boundary (`connectivity_report`). [G3](#) relaxes the gate slightly (required connected fraction 0.80 vs 0.85) and, crucially, changes the geometry — see [§4.1](#).

3 The shared motor–clutch law / 共用的馬達–離合器定律

The stochastic motor–clutch law is also inherited from [G2](#) V3 (Bell 1978; Bangasser & Odde 2013). [G3](#) applies *the identical law*, but per *protrusion* rather than to two fixed left/right ensembles. A bundle has C parallel Hookean clutches; clutch c has extension x_c and force $F_c = k_c x_c$ (stiffness $k_c = \text{clutch_stiffness}$).

Bond kinetics (Bell slip-bond). An unbound clutch binds within Δt with probability $1 - e^{-k_{\text{on}}\Delta t}$. A bound clutch’s off-rate accelerates with its own load, and it unbinds with probability $1 - e^{-k_{\text{off}}\Delta t}$:

$$k_{\text{off}} = k_{\text{off}}^0 \exp\left(\frac{F_c}{F_b}\right), \quad P_{\text{unbind}} = 1 - e^{-k_{\text{off}}\Delta t}, \quad P_{\text{bind}} = 1 - e^{-k_{\text{on}}\Delta t}, \quad (7)$$

with unloaded off-rate k_{off}^0 (`clutch_off_rate0`), on-rate k_{on} (`clutch_on_rate`), and characteristic bond force F_b (`bell_force`).

Motor force–velocity and clutch loading. The actin retrograde flow slows under load through a linear force–velocity relation, and bound clutches stretch at the difference between actin flow and substrate (fibre) motion:

$$v_{\text{actin}} = v_u \max\left(0, 1 - \frac{T}{F_{\text{stall}}}\right), \quad \dot{x}_c = \max(0, v_{\text{actin}} - v_{\text{sub}}) \quad (\text{bound}), \quad (8)$$

where $v_u = \text{unloaded_actin_speed}$, F_{stall} = the bundle’s stall force (`motor_stall_per_protrusion`), $T = \sum_{c \in \text{bound}} k_c x_c$ is the current bundle traction, and $v_{\text{sub}} = \mathbf{v}(\mathbf{x}) \cdot \hat{\mathbf{n}}_{\text{in}}$ is the inward speed of the gripped material point (its interpolated bead velocity projected onto the inward normal). The bundle traction $T \hat{\mathbf{n}}_{\text{in}}$ is applied to the two host beads of the gripped edge — the collagen is pulled *inward*, toward the cell.

The key removal. [G2](#) V3 biased the two sides with a hardcoded probability: its on-probability carried a factor $\text{side} = [2(1 - \psi), 2\psi]$ with $\psi = 0.65$ (`polarity_probability`). [G3](#) **deletes this entirely**: all sectors use the same symmetric k_{on} , and any binding bias is an *emergent* consequence of which protrusions physically reach and load fibres ([§4.3](#)).

4 What Generation-3 adds on top / G3 在其上新增的東西

The next four sections are the actual content of **G3**. They are the *only* physics that is new relative to §2–§3.

4.1 A spheroid with a fibre-free gap / 帶無纖維間隙的球體

G3 enlarges the cell into a *spheroid* of radius $R = 22\ \mu\text{m}$ (vs $9\ \mu\text{m}$ in **G2**) and leaves a fibre-free ring of width g (gap) around it. The generator rejects any candidate fibre that intrudes inside the gap radius,

$$\min_{p \in \text{fibre}} \|\mathbf{p}\| < R_g \equiv R + g \implies \text{reject the fibre}, \quad (9)$$

so at $t = 0$ there is genuinely **no cell–ECM contact**: nothing is pre-attached, and every fibre is a *potential* contact fibre rather than a pre-placed one (contrast **G2**, which seeds eight curved Bézier contact fibres reaching the surface). The spheroid must *earn* its grip by reaching across the gap.

4.2 Explicit protrusions and tip-first engagement / 顯式偽足與尖端先接觸

The membrane is divided into S equal angular sectors (`n_sectors` = 24) with outward unit normals $\hat{\mathbf{n}}_k = (\cos \theta_k, \sin \theta_k)$. Sector k carries one protrusion of length L_k , whose *tip* sits at

$$\mathbf{t}_k = \mathbf{r}_c + (R + L_k) \hat{\mathbf{n}}_k. \quad (10)$$

Length dynamics. Every sector probes outward at a baseline speed (so the spheroid explores in all directions and can break symmetry mechanically); sectors above the mean polarity grow further, sectors well below it retract, and *engaged* protrusions hold their length (they are anchored). With headroom $h_k = 1 - L_k/L_{\max}$ and normalised polarity excess $\varepsilon_k = (a_k - \bar{a})/(\bar{a} + \epsilon)$,

$$\dot{L}_k = \begin{cases} 0, & \text{engaged,} \\ \underbrace{v_{\text{probe}} h_k}_{\text{explore}} + \underbrace{v_{\text{grow}} [\varepsilon_k]_0^3 h_k}_{\text{front growth}} - \underbrace{v_{\text{ret}} [-\varepsilon_k]_0^1}_{\text{rear retraction}}, & \text{free,} \end{cases} \quad L_k \leftarrow \text{clip}(L_k + \Delta t \dot{L}_k, L_{\min}, L_{\max}), \quad (11)$$

where $[\cdot]_0^m = \text{clip}(\cdot, 0, m)$, and $v_{\text{probe}}, v_{\text{grow}}, v_{\text{ret}}$ are `protrusion_probe_speed`, `protrusion_growth_speed`, `protrusion_retraction_speed`. Reach is capped at $L_{\max} = 30\ \mu\text{m}$ (`protrusion_max_length`), i.e. roughly the gap plus a fibre or two ($\sim 0.7\text{--}4\times$ cell radius, Carey 2016; Fraley 2010/2015).

Tip-first engagement. A protrusion can bind only once its tip physically reaches a fibre. Let $d_k = \min_{\text{edges}} \text{dist}(\mathbf{t}_k, \text{edge})$ be the distance from the tip to the nearest material point on any fibre segment (with foot-point parameter α_k and host edge e_k). Then

$$d_k \leq d_{\text{cap}} \implies \text{sector } k \text{ engages edge } e_k \text{ at } \alpha_k, \quad (12)$$

with capture distance d_{cap} (`capture_distance` = $2.5\ \mu\text{m}$). An engaged sector stays engaged while any of its clutches is bound; when all release, its tip is re-tested against the (now moved) fibre and it disengages if $d_k > d_{\text{cap}}$. Engagement is refreshed every `engagement_update_interval` = 0.5 s.

4.3 Emergent polarity: wave-pinning + adhesion feedback / 湧現極性

This is the heart of the rebuild and the direct replacement for the deleted `polarity_probability` = 0.65. The membrane carries a scalar *activity* field a_k ($k = 1, \dots, S$), a coarse-grained Rho-GTPase-like “front” marker, subject to a **mass-conserved replicator** / **wave-pinning** dynamic (Mori, Jilkine & Edelstein-Keshet 2008). The conserved pool is

$$\sum_{k=1}^S a_k = A_{\text{tot}} \quad (\text{polarity_total_activity}), \quad (13)$$

enforced by renormalisation each step (this is the *global inhibition*). Local self-reinforcement is a saturating Hill term plus the FAK/Rac1 adhesion signal q_k (§4.4):

$$\phi_k = \gamma \frac{a_k^h}{K^h + a_k^h} + k_{\text{FAK}} q_k, \quad \text{fitness}_k = \phi_k - \langle \phi \rangle, \quad (14)$$

where $\gamma = \text{polarity_gamma}$, $K = \text{polarity_K}$, $h = \text{polarity_hill}$, $k_{\text{FAK}} = \text{polarity_fak_gain}$, and $\langle \phi \rangle$ is the sector mean. The zero-sum fitness makes it a **replicator** (winner-take-all): sectors whose reinforcement beats the mean grow at the expense of the rest. Membrane diffusion (a discrete ring Laplacian) keeps the front contiguous, and a small stochastic term breaks the initial symmetry:

$$(\text{lap } a)_k = a_{k+1} + a_{k-1} - 2a_k, \quad a_k \leftarrow a_k + \frac{\Delta t}{\tau_p} (a_k \text{fitness}_k + D (\text{lap } a)_k) + \sigma \sqrt{\Delta t} \xi_k, \quad (15)$$

followed by $a_k \leftarrow \max(a_k, 10^{-4})$ and renormalisation to Eq. (13). Here $\tau_p = \text{polarity_time}$ (front relaxation time), $D = \text{polarity_diffusion}$, $\sigma = \text{polarity_noise}$, and $\xi_k \sim \mathcal{N}(0, 1)$ i.i.d. per step.

How it works. From a near-uniform, symmetric start ($a_k \approx A_{\text{tot}}/S$ with 2% noise), the bistable local kinetics plus mass conservation collapse the field to **one stable broad front** (a pinned wave), rather than a spread of many peaks. Three ingredients play distinct roles:

- **Noise** $\sigma \sqrt{\Delta t} \xi_k$ chooses *which* direction — the sign is genuinely stochastic and differs between seeds (verified: five seeds polarise toward 84° , -84° , -8° , -99° , 64°).
- **Adhesion feedback** $k_{\text{FAK}} q_k$ pulls the front toward sectors that *actually grip* collagen — a mechanochemical positive loop.
- **Aligned collagen** (optional fixture) biases only the front *axis*, never its sign.

Nothing prescribes $+x$, a left/right probability, or the number 0.65.

4.4 FAK/Rac1 adhesion signal / 黏著訊號

The feedback variable q_k is a low-pass-filtered measure of adhesion success in sector k . Let n_k be the number of currently bound clutches and T_k the bundle traction. Define an engagement saturation and a normalised load, combine them, and filter:

$$q_k^{\text{raw}} = \underbrace{(1 - e^{-n_k/n_s})}_{\text{engagement}} \cdot \underbrace{\frac{\lambda_k}{1 + \lambda_k}}_{\text{load}}, \quad \lambda_k = \frac{T_k}{\frac{1}{2} F_{\text{stall}}}, \quad q_k \leftarrow q_k + \frac{\Delta t}{\tau_q} (q_k^{\text{raw}} - q_k), \quad (16)$$

with clutch scale $n_s = \text{adhesion_clutch_scale}$ and filter time $\tau_q = \text{adhesion_filter_time}$. The low-pass ($\tau_q = 20$ s) makes the feedback respond to *sustained* traction rather than momentary binding flicker, which is what stabilises a single front instead of the “twitchy” flicker of the superseded **G3**.

4.5 Reaction-driven 2-D migration / 反作用力驅動的二維遷移

Motion follows the same reaction principle as [G2 V3](#) but in full 2-D. The clutch bundles apply a total inward force field $\mathbf{F}^{\text{active}}$ on the collagen ([§3](#)). Because those clutch forces are internal to the cell–ECM pair, the net reaction on the rigid spheroid is minus their sum, and it points *toward* the gripping front:

$$\mathbf{R} = - \sum_i \mathbf{F}_i^{\text{active}}, \quad \mathbf{v}_c = \frac{\mathbf{R}}{\gamma_c}, \quad \mathbf{v}_c \leftarrow \mathbf{v}_c \frac{\min(\|\mathbf{v}_c\|, v_{\max})}{\|\mathbf{v}_c\|}, \quad \mathbf{r}_c \leftarrow \mathbf{r}_c + \Delta t \mathbf{v}_c, \quad (17)$$

with cell drag $\gamma_c = \text{cell_drag} = 3000 \text{ nNs}/\mu\text{m}$ (calibrated to $0.1\text{--}0.3 \mu\text{m}/\text{min}$) and a numerical/biological speed guard $v_{\max} = \text{max_cell_speed}$. In `--fixed` mode the spheroid is clamped ($\mathbf{v}_c \equiv 0$): fibres still get pulled and realigned, but there is no migration — a clean falsifiability control.

Contrast with [G2 V3](#): V3 intentionally allowed only one translational degree of freedom, $\dot{x}_c = \text{reaction}_x/\gamma_c$, $\dot{y}_c = 0$, on a declared left/right axis. [G3](#) has no declared axis: both components of $\mathbf{R}$ act, and the direction is set by the emergent front.

5 The full G3 time step / G3 完整時間步

Each step of `run_spheroid` (size Δt) executes the following, in order. Steps 1–3 and 5 are new; step 4 is the shared clutch law ([§3](#)); step 7 is the shared engine ([§2](#)).

1. **Emergent polarity** — one wave-pinning update, Eq. ([15](#)) (`step_polarity`).
2. **Protrusion growth** — update every L_k , Eq. ([11](#)) (`step_protrusions`).
3. **Tip-first engagement** — every 0.5 s, re-test tips and (dis)engage, Eq. ([12](#)) (`update_engagement`).
4. **Motor–clutch loading** — advance every bundle, build the inward nodal force field $\mathbf{F}^{\text{active}}$, Eqs. ([7](#))–([8](#)) (`step_clutches`).
5. **Adhesion filter** — update q_k , Eq. ([16](#)) (`update_adhesion`).
6. **Reaction-driven motion** — move the spheroid, Eq. ([17](#)).
7. **Advance the collagen** — one overdamped step of the frozen [G2](#) engine, Eqs. ([1](#))–([6](#)) (`fast_advance`); returns the bead velocity field used as v_{sub} next step.

The narrative this produces: $t = 0$ fibre-free gap $\rightarrow$ protrusions probe outward in all directions $\rightarrow$ some tips reach fibres and bind $\rightarrow$ motor–clutches load and pull those fibres inward (the network realigns radially at the front) $\rightarrow$ a single front is selected and reinforced by adhesion feedback $\rightarrow$ the spheroid migrates toward that front, remodelling collagen along its path.

6 Metrics / 量測

Radial nematic order. For each fibre segment with unit tangent $\hat{\mathbf{t}}$ and midpoint radial unit vector $\hat{\mathbf{e}}_r$ (from the spheroid centre), the nematic radial order is

$$S_r = \langle 2(\hat{\mathbf{t}} \cdot \hat{\mathbf{e}}_r)^2 - 1 \rangle, \quad (18)$$

averaged in near/mid/far shells (surface distances $0\text{--}20$, $20\text{--}45$, $45\text{--}90 \mu\text{m}$). $S_r = +1$ means fibres point radially (toward the spheroid), 0 random, -1 tangential. Radial pulling drives the near-shell S_r from ≈ -0.62 toward -0.31 over the run.

Polarity vector and traction. The net polarity direction is $\mathbf{P} = \sum_k a_k \hat{\mathbf{n}}_k$ (its angle is the migration heading); the collective traction is $\sum_k T_k$; and the path length accumulates $\|\mathbf{r}_c(t + \Delta t) - \mathbf{r}_c(t)\|$.

7 Parameters and literature anchors / 參數與文獻依據

Symbol	Config field	Value	Meaning / source
<i>Shared G2 engine</i>			
L	domain_size	240 μm	field size (G2: 180)
E	collagen_modulus_mpa	32 MPa	wet fibre modulus (Lee 2014)
D	fiber_diameter	0.30 μm	fibre diameter
ρ	compression_ratio	0.10	microbuckling softening (Notbohm 2015)
k_x	crosslink_stiffness	75 nN/ μm	hinged-link penalty
k_{rep}	repulsion_stiffness	600 nN/ μm	cell-surface penalty
ζ	bead_drag	360 nN s/ μm	bead drag (G2 V3)
Δt	dt	0.05 s	integration step
<i>Spheroid & protrusions (new)</i>			
R	cell_radius	22 μm	spheroid radius (G2: 9)
g	gap	12 μm	fibre-free ring
N_f	n_fibers	150	fibre count (G2: 99)
S	n_sectors	24	membrane sectors / protrusions
L_{max}	protrusion_max_length	30 μm	max reach (Carey 2016)
v_{probe}	protrusion_probe_speed	0.14 $\mu\text{m/s}$	baseline probing
v_{grow}	protrusion_growth_speed	0.16 $\mu\text{m/s}$	front growth
v_{ret}	protrusion_retraction_speed	0.22 $\mu\text{m/s}$	rear retraction
d_{cap}	capture_distance	2.5 μm	tip capture radius
<i>Motor-clutch bundle (shared law)</i>			
C	n_clutches_per_protrusion	8	clutches per bundle
k_c	clutch_stiffness	2.0 nN/ μm	clutch spring
k_{on}	clutch_on_rate	0.08 s ⁻¹	symmetric binding
k_{off}^0	clutch_off_rate0	0.02 s ⁻¹	unloaded off-rate
F_b	bell_force	2.0 nN	Bell bond force
v_u	unloaded_actin_speed	0.03 $\mu\text{m/s}$	unloaded actin flow
F_{stall}	motor_stall_per_protrusion	4.0 nN	motor stall
<i>Emergent polarity + adhesion (new)</i>			
A_{tot}	polarity_total_activity	6.0	conserved pool
γ	polarity_gamma	2.2	autocatalytic gain
K	polarity_K	0.5	Hill half-saturation
h	polarity_hill	2.0	Hill cooperativity
D	polarity_diffusion	0.45	membrane smoothing
σ	polarity_noise	0.08	symmetry-breaking noise
k_{FAK}	polarity_fak_gain	2.5	FAK/Rac1 feedback gain
τ_p	polarity_time	6.0 s	front relaxation time
τ_q	adhesion_filter_time	20 s	q_k low-pass
n_s	adhesion_clutch_scale	3.0	engagement scale
<i>Motion (new geometry, shared principle)</i>			
γ_c	cell_drag	3000 nN s/ μm	spheroid drag

Symbol	Config field	Value	Meaning / source
$v_{\max}$	<code>max_cell_speed</code>	$0.02 \mu\text{m/s}$	speed guard

The literature sets *targets and plausible scales*, not fits: spheroid traction of tens of nN (Steinwachs 2016; Mark 2020), MDA-MB-231 3-D speed $0.1\text{--}0.3 \mu\text{m/min}$ (Sapudom 2019; Aguilar-Rojas 2024), protrusion reach $\sim 0.7\text{--}4 \times$ cell radius (Carey 2016; Fraley 2010/2015), wave-pinning polarisation in ~ 10 s with stochastic sign (Mori 2008; Jilkin 2011), and fibre realignment over tens of minutes to hours (Kim 2017; Han 2018).

8 Side-by-side: G2 vs G3 / 逐項對照

Aspect	G2 (corrected)	G3 (spheroid guidance)
Collagen engine	bead-spring, overdamped, axial EA/ℓ_0 , bending EI/ℓ^3 , hinged crosslinks, boundary anchoring, soft repulsion	identical — reused verbatim (§2)
Integrator	<code>Network.advance</code> (np.add.at)	<code>fast_advance</code> (np.bincount); bit-identical , $\sim 2.4 \times$ faster
Cell body	rigid circle, $R = 9 \mu\text{m}$	rigid spheroid, $R = 22 \mu\text{m}$
Contact at $t = 0$	8 pre-placed Bézier contact fibres touch the surface	fibre-free gap — no contact; nothing pre-attached (Eq. 9)
Cell-ECM coupling	Gaussian-weighted contact patches in a $\pm 30^\circ$ sector	explicit protrusions , tip-first engagement (Eqs. 10–12)
Motor-clutch law	Bell slip-bond + motor F–V (V3)	same law , one bundle per protrusion (§3)
Directional bias	hardcoded <code>polarity_probability=</code> (V3 side factor) 0.65	removed ; emergent wave-pinning + FAK feedback (§4.3)
Polarity origin	prescribed constant	noise breaks symmetry; mechanics pick direction; sign is stochastic per seed
Migration	1-D, $\dot{x}_c = \text{reaction}_x/\gamma_c$, $\dot{y}_c = 0$ (V3)	full 2-D , $\mathbf{v}_c = \mathbf{R}/\gamma_c$ (Eq. 17)
Force scale	5 nN local pull (V2)	tens of nN collective (~ 36 nN peak)
Domain / fibres	$180 \mu\text{m}$ / 99 fibres	$240 \mu\text{m}$ / 150 fibres
Percolation gate	connected fraction ≥ 0.85	≥ 0.80 (relaxed for the gap)
Duration	30–60 min	60 min (realign + migrate)
Constitutive law	purely elastic; no SLS, no plasticity	identical — no SLS, no plasticity

One-line summary. G3 = the G2 collagen engine and G2 motor-clutch law, unchanged, wrapped around a spheroid-with-gap geometry, explicit tip-first protrusions, an emergent wave-pinning polarity field that *replaces* the 0.65 constant, and full-2-D reaction-driven motion.

9 Reference run and falsifiability / 參考結果與可證偽性

The reference run — 150 fibres, 240 μm field, 22 μm spheroid, 12 μm gap, 60 min, seed 7 (results/g3_spheroid_guidance/main/) — gives:

Quantity	Value	Literature target
Peak collective traction	$\approx 36 \text{ nN}$	tens of nN, ≤ 100 (Steinwachs 2016)
Net migration in 60 min	$\approx 18.8 \mu\text{m}$	—
Migration speed	$\approx 0.32 \mu\text{m}/\text{min}$	0.1–0.3 $\mu\text{m}/\text{min}$ (Sapudom 2019)
Engaged protrusions at end	≈ 10	—
Near-shell radial order	$-0.62 \rightarrow -0.31$	realignment over tens of min–h
Max bead displacement	$\approx 5.1 \mu\text{m}$	—
Connected fraction	1.0	gate ≥ 0.80

Emergence check. Across five seeds the migration heading is 84° , -84° , -8° , -99° , 64° (spread $\approx 75^\circ$): the direction is genuinely stochastic and set per realisation, confirming there is no built-in bias.

Controls that make the mechanism falsifiable.

- `--fixed` clamps the spheroid: fibres still realign, but migration vanishes.
- Zero motor stall force: protrusions attach without deforming fibres.
- An empty gap with no reachable fibre: zero engagement, zero traction, zero motion.

Two view modes are rendered in the **G2** visual grammar (springs coloured by strain, gold-diamond crosslinks, black-square fixed beads, peach cell, orange protrusions/contacts, faint ghost of the initial network, true $1\times$ scale + scale bar): `*_full.gif` (full field) and `*_follow.gif` (follow-cell zoom), matching the G2-v3 notebook toggle.

10 Evidence boundary / 證據邊界

G3 is a **2-D minimal mechanism model of emergent cell–collagen guidance**. It is *not* a realistic three-dimensional tumour-migration prediction: it contains no calibrated collagen concentration or pore topology, no deformable cell or nucleus, no proteolysis, no EMT, no cell–cell interactions, no stress relaxation, and no irreversible plasticity. **G2** (generations/g2_corrected/) remains frozen and unchanged; **G3** reuses its engine without rewriting any **G2** file. Every generated figure carries the boundary: “*Mechanism demo — not a 3-D tumour-migration prediction.*”

References / 參考文獻

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Source of truth: `generations/g3_spheroid_guidance/model.py` (`SpheroidConfig`, `step_polarity`, `step_protrusions`, `update_engagement`, `step_clutches`, `update_adhesion`, `run_spheroid`, `fast_advance`) and the frozen [G2](#) engine `generations/g2_corrected/common/model.py`. All equations above were transcribed directly from that code.